

Digital Human Live Commerce Case Study

Unmanned, Script-Driven AI Livestreaming at Scale

Note: This case is written *qualitatively* — the digital-human project's internal result metrics (PV/CVR/GMV) were never publicly disclosed, so it is framed around the 0-to-1 capability, the reliability-first judgement and the commercial use case rather than performance numbers.

One-line judgement: For live commerce, the bottleneck was never content — it was the cost and availability of a human anchor. The winning product was not a more lifelike avatar, but an unmanned, script-driven live room that a non-technical merchant could set up and trust to run 24/7.



Role & Scope

Role: Lead Product Manager (owned the product end-to-end)

Product: Digital Human Live Commerce — an AI digital-human livestreaming platform that lets merchants run automated, always-on live selling using an established AI persona.

Scope: Problem framing, product strategy, B-end (merchant setup) and C-end (viewer experience) design, real-time interaction model, applied-AI scoping, cross-functional delivery with engineering, algorithm, design and operations.

Context: Livestream commerce boom + pandemic-era pressure on offline retail; The firm was building out its live-commerce platform (a £2M+ daily GMV business) and looking to give an established AI persona a real commercial landing.

Context

Livestream shopping had become one of the fastest-growing commerce formats, but it ran on a fragile, expensive input: **human anchors**. A live room needs a person who is on-camera, on-script and on-brand — which means real anchors are costly, hard to schedule, impossible to run around the clock, and out of reach for small and micro merchants. During the pandemic this gap widened: many merchants, and even farmers selling produce, needed to sell online but had no anchor, no studio and no budget.

At the same time, the firm had an established AI persona that needed commercial, revenue-generating use cases rather than only marketing moments.

The initial framing was tempting but wrong: *"build a more realistic AI streamer."* I reframed it around the real constraint.

Problem

The core problem was not *"livestream rooms need better-looking hosts."* It was:

"How does a merchant with no anchor, no studio and no budget run a credible live room that sells — reliably, and around the clock?"

That reframes the product from an **avatar-quality problem** into an **operating-cost and reliability problem**. The AI digital human is only valuable if it removes the human bottleneck *without* making the live room feel broken, off-brand or untrustworthy to shoppers.

Insight

Merchants don't want a magical AI. They want a live room that **runs without them, stays on-brand, and doesn't embarrass them in front of customers**.

That shifted the design goal from "impressive generation" to **"reliable, controllable, merchant-operable automation"** — an AI product judged by uptime, brand-safety and conversion, not by how lifelike a single clip looks.

Strategic Decisions

1. **Script-driven, not free-form.** Rather than let a generative model improvise the whole broadcast, I structured the live room around configurable **script modules** — opening/closing, marketing, product explanation, fixed interactions, and real-time interactions. A merchant edits and publishes these before going live, and the system renders them into a continuous audio-video stream. This traded spontaneity for **reliability, brand control and merchant setup without any technical skill** — the qualities a commerce product actually needs.

2. **A real-time interaction engine on top of the script.** To avoid a "dead" pre-rendered feel, the live room reacts to viewer behaviour — entry, likes, follows, gifts, and orders each trigger the matching bound action clip plus text-to-speech, composed live into the stream (e.g. "Welcome, [user] — thanks for the like!"). This kept the room feeling live and responsive while staying fully automated.

3. **Ship the answerable version of Q&A; first.** True real-time semantic Q&A; (voice + matching body language) was the ideal, but the highest-risk piece. I scoped Phase 1 to fall back to a **text-bubble reply** when full voice-action response wasn't ready — so the product shipped and sold, instead of blocking on the hardest AI problem.

4. **Leverage an existing, trusted IP (an established AI persona).** Rather than a generic avatar, binding the room to a persona users already recognised gave it warmth, personality and stickiness — and gave the firm's AI persona a genuine commercial home.

Trade-offs

The central trade-off was **control vs. spontaneity**.

A script-driven digital human is less "wow" than a fully generative one — it won't freestyle. But in a commerce setting, freestyling is a liability: it risks off-brand claims, wrong product details and moments no merchant would sign off on. Choosing a controllable, configurable system meant the product was **trustworthy enough to actually sell under a brand's name**, and simple enough that a non-technical merchant could operate it. The bet was that reliability and brand-safety would matter more than novelty for real GMV — which is the judgement I'd defend.

The second trade-off was **automation vs. human warmth**. Fully unmanned live risks feeling robotic; the real-time interaction engine and a familiar persona were the deliberate counter-weights.

Outcomes & Impact

- Built the firm's digital-human live-commerce capability **0-to-1** — a script-driven, fully unmanned live room covering the whole selling loop: opening, product explanation, lucky-draws, coupons, real-time interaction and Q&A.;
- Enabled **24/7 unmanned live selling**, removing the anchor, studio and manpower cost that previously gated a live room — extending live commerce to merchants who couldn't staff an anchor, including small/micro and agricultural sellers.
- Landed the **established brand-IP digital human** as a commercial, revenue-generating selling scenario, not just a marketing moment — connecting it with quality supply and digital-collectible tie-ins.
- Defined the **"24H stand-in anchor"** and **"human + digital-human co-hosting"** models from competitive research (benchmarking comparable products), giving the platform a differentiated path.

What I Learned

- **In commerce AI, controllability beats spontaneity.** The product succeeded because merchants could trust and operate it — not because the avatar was the most lifelike. Judging AI features by uptime, brand-safety and conversion (not by demo wow-factor) is the discipline that shipped it.
- **Apply AI where it removes a real cost, not for novelty.** The value wasn't "an AI streamer" — it was deleting the human bottleneck that kept live commerce out of most merchants' reach.
- **Meet non-technical users where they are.** Turning a hard AI capability into a *configurable script* a merchant could set up themselves is what made the automation actually get adopted.
- **Scope around the riskiest assumption.** Falling back to text-bubble Q&A; in Phase 1 let the product ship and learn, instead of stalling on the hardest interaction problem.